

## one small radiomics cohort

$n_+ = 57$  events,  $n_- = 126$ , 1004 features

honest  
nested CV

### HONEST EVALUATION

**0.46**

$\approx$  chance  
(leakage-free)

shop the best  
of many splits

### REPORTED EVALUATION

**0.68**

looks like signal  
(typical split: 0.54)

### selection optimism

$$0.68 - 0.54 = 0.14$$



**this gap is sampling variance, set by class counts,  
not dimensionality; predictable pre-modeling:**

$$\Delta \approx K \cdot \text{SE}_{\text{AUC}}(n_+, n_-)$$

**closed-form SE**  $R^2=0.67$  vs **dimensionality**  $R^2=0.09$

*across 50 cohorts, from class counts alone; 0.64 predicting a held-out source.  
The higher 0.82 evaluates the SE at each cohort's own honest AUC, so it needs a model first.*